Introduction

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Nonferrous powder production methods are multifarious and numerous. These techniques allow the manufacture of a wide spectrum of metal powders designed to meet the requirements of a variety of applications. Various powder production processes allow precise control of the powders' chemical composition and physical characteristics to obtain prescribed attributes for the features of different applications.

The ratio of nonferrous powder to ferrous powder shipments (expressed in tonnage) is about one to four. But in market value terms, aluminum, silver, and zinc are all close to the value of iron powders [1, 2]. This is illustrated by data about worldwide yields of metal powder manufactured by all processes [2]. Fig. 1 shows the price in \$/kg of powder, its volume in t/yr, and its total market value in millions of dollars (log scale).

BASIC METHODS OF POWDER PRODUCTION

Nonferrous powders are produced by mechanical, chemical, and electrochemical methods. Mechanical methods include the comminution of solids and atomization of melts. The comminution of solid metals and alloys is carried out by machining, milling (in impact mills, ball mills, vibration mills, stream mills, and edge-runner mills), and using attritors. General chemical methods include reduction methods, precipitation from solution, hydrometallurgical processing in leach autoclaves, the carbonyl process, and the hydride/dehydride process. Electrolytic methods include electrolysis of aqueous solutions, melt electrolysis, and the cementing process (Table 1).

The shape and size of particles depend on the method of powder production and may vary widely. In Fig. 2, common particle shapes produced by different methods as depicted in ISO 3252 are shown.

Atomization is the most universal method used that allows powder production over a wide range of compositions and in a wide variety of particle sizes, from 1 μm to a few mm. While ferrous powders are currently produced by the water atomization of liquid steel, nonferrous powder production methods are multifarious and numerous. Chemical and electrolytic methods are widely used for producing high-purity nonferrous powders. One basic nanopowder production method is extraction from the gaseous phase.

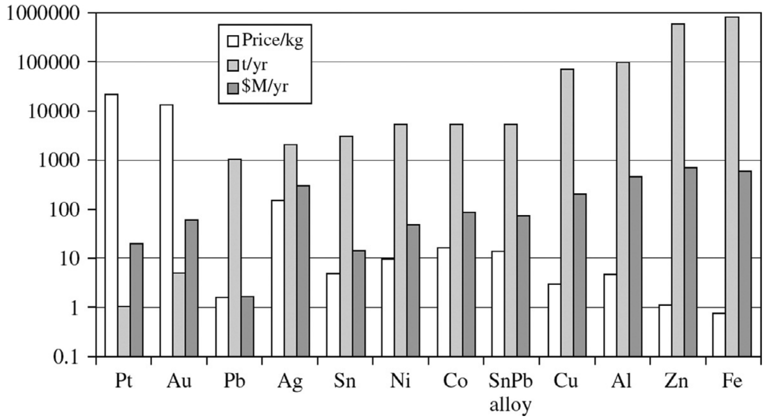

FIG. 1 Yield and value of metal powders.

TABLE 1 Basic Methods of Non-Ferrous Metals Powder Production

Method of Production	Processing Character	Material	Characteristic of Powder Particles: Mean Mass Diameter (δ_m), μm	Characteristic of Powder Particles: Shape
Mechanical methods				
Dispersion of solids	Machining	Brass, bronze, Mg, Al, Cu	3×10^3–10^3	Chips
	Hammer mills	Sinter cakes	5×10^3–50	Angular, irregular
	Ball, rod, and pebble mills, self-grinding	Bronze, Al, Ag, Ti, Ni, Cr	5×10^3–10	Scaly, flake
	Mill rolls and bowl mills	The same	10^4–20	Irregular, scaly
	Vibratory mills	Metals, alloys	10–0.2	Scaly
	Attritors	Metals, alloys	15–0.1	Scaly
	Stream mills	Mo, Be, tungsten alloys	40–1.0	Irregular
Dispersion of melts	Water granulation	Metals, alloys	5×10^3–10^3	Spherical, irregular
	Water atomization	Metals, alloys	2×10^3–1.0	Irregular
	Gas atomization	Metals, alloys	500–10	Spherical, irregular
	Centrifugal atomization	Al alloys, Be alloys, Mg, Mo, Ni alloys	150–50	Spherical, irregular
	Rotating electrode process	Ti alloys, Ni alloys	300–150	Spheroidal Irregular, angular
Chemical and electrolytic methods				
Reduction metal oxides	Oxide reduction by reducing atmospheres	Ni, Co, Cu, W, Mo	300–20	Spongy, granular, (conglomerates)
	Precipitation from solution	Ti, W, Precious metals	100–1, nano size	Irregular, angular
	Hydrogenation, dehydrogenation	Ti, Zr, Hf	200–20	Spongy, irregular, angular
	Hydrometallurgical processing in leach autoclaves	Ni, Co, Cu, Precious metals	70–10	Spongy, granular, briquettes
	Metallothermy	Tantalum, niobium, magnesium, titanium, thorium, zirconium, vanadium	300–20	Acicular, spongy
	Carbothermy	Aluminum, magnesium, iron, niobium, tungsten	400–0.1	Angular, nodular
Electrolysis	Electrolysis of aqueous solutions	Copper, tin, lead, zinc, nickel	400–0.1	Dendritic
	Electrolysis from melts	Aluminum, magnesium, titanium, tantalum, niobium, vanadium, zirconium	400–10	Dendritic
Physical-chemical methods				
Gas phase precipitation	Evaporation and condensation	Zn, Mg, Ti, Ag, Cd, Bi	Ultrafine, Nano size	Conglomerate of particles, clusters
Carbonyl-process	Chemical precipitation, evaporation and codensation	Ni, Co, W, Mo	300–2	Acicular, granular

TABLE 1 Basic Methods of Non-Ferrous Metals Powder Production—cont'd

Method of Production	Processing Character	Material	Characteristic of Powder Particles	
			Mean Mass Diameter (δ_m), μm	Shape
Other methods				
Mechanical alloying	Ball milling	Metals, alloys	Ultrafine, nano size	Agglomerate of particles, clusters
Spray granulation	Fluidized bed spray granulator	Al, Co, Pyrophoric Rare Earths	5×10^3–10^3	Spherical

Comminution of most solid metals is significantly limited due to their ductility. Nevertheless, milling in high-energy apparatus, such as attritors, finds wide application for mechanically alloying powders. Some basic physical and chemical methods of nanopowder and ultrafine powder production are discussed in Chapters 3, 5, and 9.

Mechanical alloying is a simple and efficacious technique for synthesizing both equilibrium and nonequilibrium phases of commercially and scientifically advantageous materials obtained from initial elemental powders. The exclusive advantages of mechanical alloying consist in the possibility of the synthesis of unique alloys that are not accessible by other technique, because mechanical alloying is a completely solid-state process and, therefore, limitations specified by state diagram do not request here. Mechanical alloying is realized by means of high-energy ball milling technique.

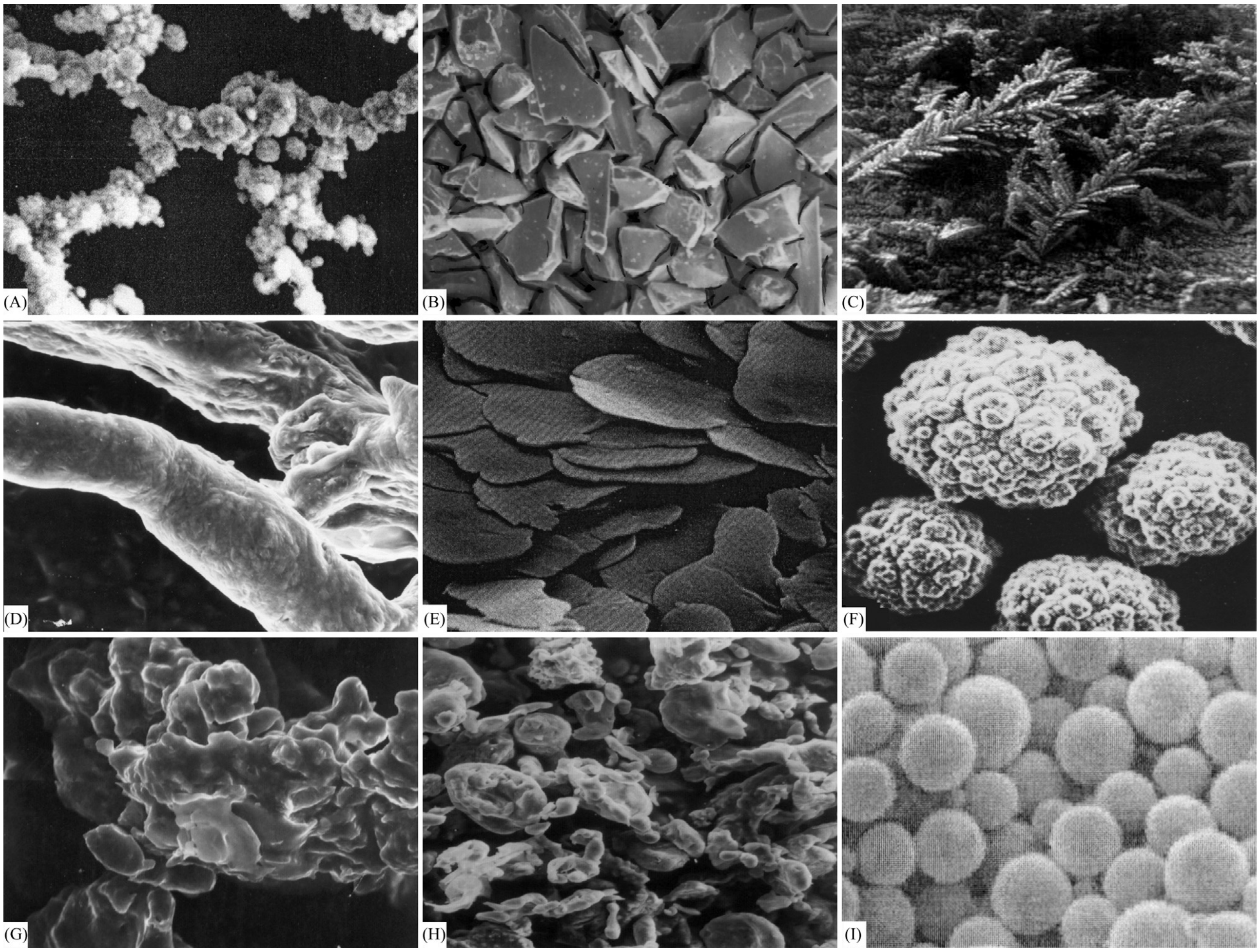

FIG. 2 Characteristic particle shapes: (A) Acicular powder particles; (B) Angular powder particles; (C) Dendritic powder particles; (D) Fibrous powder particles; (E) Flaky powder particles; (F) Granular powder particles; (G) Irregular powder particles; (H) Nodular powder particles; (I) Spheroidal powder particles.

Water-atomized powder-based aluminum alloys with nanoquasicrystalline phases display high mechanical properties and heat-resistance at both room and elevated temperatures [3]. Their high strength results from the high hardness and high Young's modulus of alloys with quasicrystalline phases. Low thermal conductivity combines herein with a thermal expansion coefficient approximate to the thermal expansion coefficient for nickel-, aluminum-and iron-based structural materials.

Additive manufacturing (AM) technologies fabricate models by fusing, sintering, or polymerizing materials in predetermined layers without tools. AM enables the manufacture of complex geometries, including detailed internal parts that are essentially impossible to manufacture using machining and molding processes, because AM does not require predetermined tool paths, draft angles, under cuts. [4] In AM, a model's layers are formed by combining computer-aided design (CAD) data with professional software. In powder-bed-based additive manufacturing, the choice of powder is crucial for the performance and cost of the built parts, for a number of reasons. In general, the powders require high apparent densities to achieve maximum density when deposited in packed beds. Chemical composition and the presence of impurities in the powder translate into the final material and its properties. The particle size distribution and morphology determine the powder's flowability and packing density. Chapter 13 contains discussion of the aspects of powders for additive manufacturing processes.

Grapheme oxide composite powders offer the perfect layer structure of two-dimensional (2D) sp2-hybridized carbon atoms; graphene has been extensively investigated in recent years due to its outstanding properties, such as high Young's modulus, high fracture strength, and excellent thermal conductivity [5]. Due to these excellent mechanical properties and high specific surface area, graphene is expected to be an ideal reinforcement phase interacting with a metal matrix, even in a small concentration. The production technique using grapheme oxide composite powders is considered in Chapter 12.

Powder particle size is a dominant characteristic of a metal powder designated for subsequent use in PM. The application of the powder determines the required size range. In Fig. 3, the length of the segment lines approximately shows these application regions in terms of powder size. A rising number of PM applications require submicron powders produced by chemical and physical methods. Powder applications embraces particle sizes ranging from a few nanometers to granules millimeters in size. The main problems in the production and processing of metallic nanopowders are caused by their high reactivity, explosibility (due to high specific area), and agglomeration of particles. In Chapter 9, "Nanopowders," a novel method for manufacturing semi-products directly from metal vapor, omitting the production of nanopowder in powdered form, is discussed.

Powder characterization and testing described in the Chapter 1 is key to the evaluation of powder production, and precedes the description and discussion of powder manufacturing methods. The main test methods and specific testing

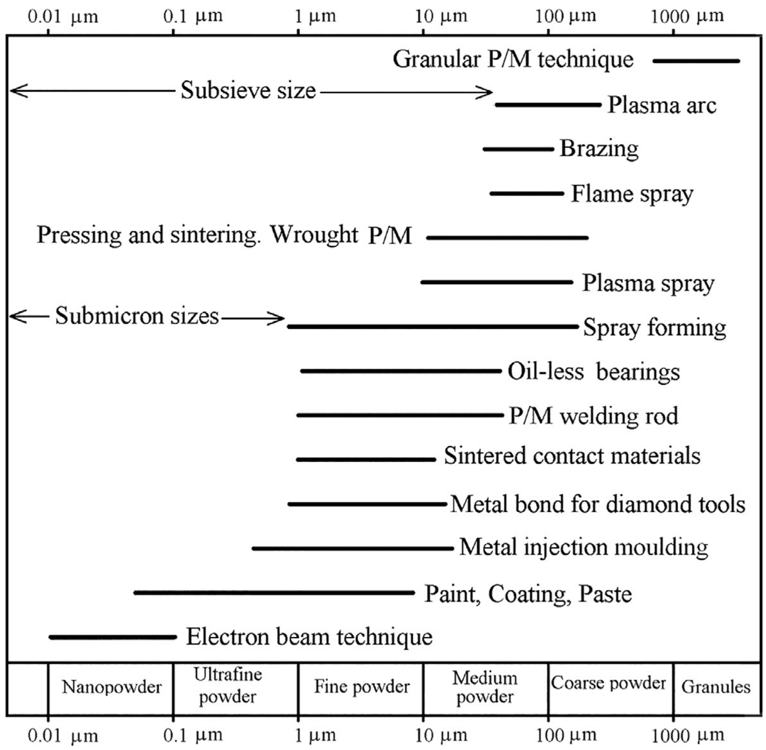

FIG. 3 Ranges of particle size suitable for different applications of non-ferrous powders.

procedures that characterize powders' physical properties, such as particle size distribution, particle shape, and surface area, are shown. Each of these parameters may have an important effect on the physical and mechanical behavior of powders produced for manufacturing PM products. Bulk properties of powders are also examined in this chapter. Finally, test methods for determination of compressibility (a key factor in reaching high part density) are given.

The fundamentals of the main methods of nonferrous metal powder production, including mechanical, chemical, electrolytic, and physical–chemical methods, are described and discussed in Section B.

Chapter 14 describes techniques used for secondary operations such as dehydrogenation, powder passivation, powder classification according to size, mixing, dosage, and packing.

Production methods and techniques for aluminum and aluminum-alloy powders, magnesium and magnesium-based powders, titanium and titanium-alloy powders, copper and copper-based powders, zinc, cadmium, and their alloy powders, nickel, cobalt, and their alloy powders, refractory metal powders, rare metal powders, noble metal powders, and many other nonferrous metal powders are described and discussed in Section C.

Chapter 27 contains the description and discussion of production safety engineering techniques and environmental protection.

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